Diagramas UML del Sistema de Mapas

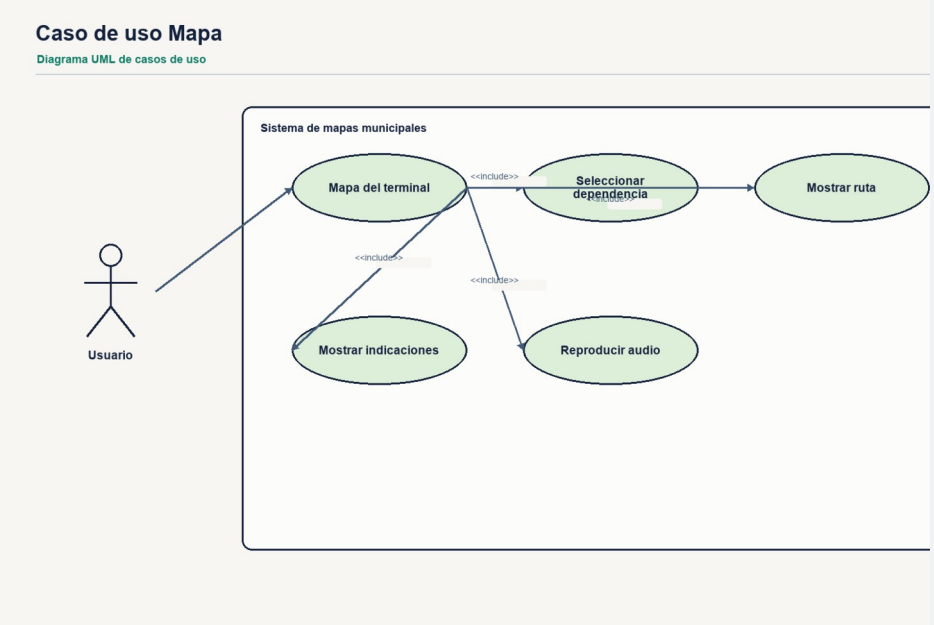
Mapa MapaR MapaConf y MapaRConf

Cada sección incluye una imagen generada desde PlantUML, la explicación del flujo y el código fuente UML correspondiente. Los archivos individuales se entregan en la carpeta plantuml.

Casos de uso

1. Caso de uso Mapa

Muestra cómo un visitante consulta una dependencia del terminal: selecciona la oficina, observa el recorrido, lee las indicaciones y puede escuchar el audio asociado.



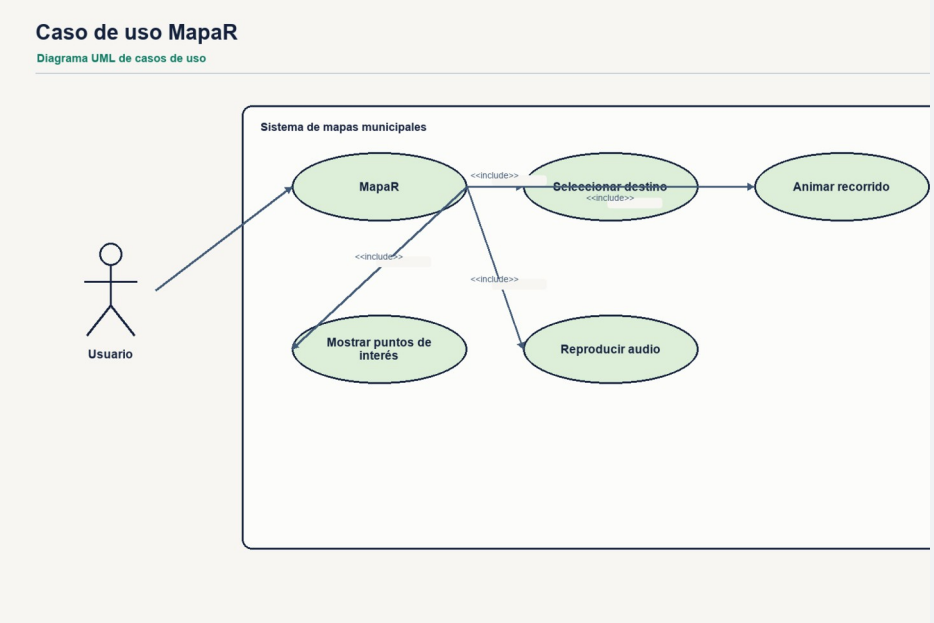

```
@startuml
title Caso de uso Mapa
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

left to right direction
actor Usuario
rectangle "Mapa del terminal" {
  usecase "Seleccionar
dependencia" as UC1
  usecase "Mostrar ruta" as UC2
  usecase "Mostrar
indicaciones" as UC3
  usecase "Reproducir audio" as UC4
}
Usuario --> UC1
UC1 .> UC2 : <<include>>
UC2 .> UC3 : <<include>>
UC2 .> UC4 : <<extend>>
@enduml
```

Figura 1. Caso de uso Mapa

2. Caso de uso MapaR

Describe la consulta de un destino urbano. El usuario escoge un destino y el sistema anima el recorrido, presenta puntos de interés, instrucciones y audio.



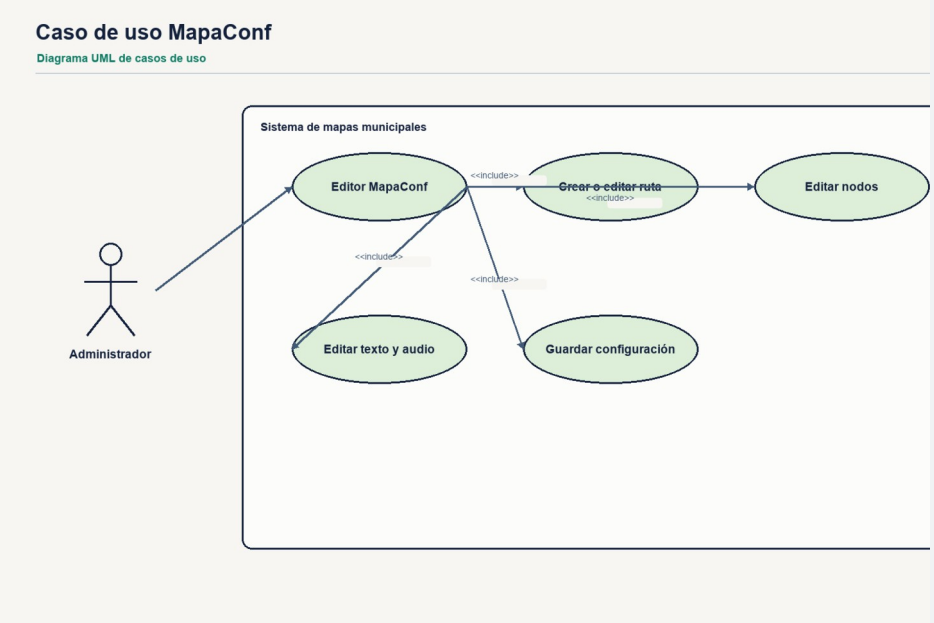

```
@startuml
title Caso de uso MapaR
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

left to right direction
actor Usuario
rectangle "Mapa urbano de Retalhuleu" {
  usecase "Seleccionar destino" as UC1
  usecase "Animar recorrido" as UC2
  usecase "Mostrar puntos
de interés" as UC3
  usecase "Mostrar
indicaciones" as UC4
  usecase "Reproducir audio" as UC5
}
Usuario --> UC1
UC1 .> UC2 : <<include>>
UC2 .> UC3 : <<include>>
UC2 .> UC4 : <<include>>
UC4 .> UC5 : <<extend>>
@enduml
```

Figura 2. Caso de uso MapaR

3. Caso de uso MapaConf

Representa las tareas del administrador en el editor del terminal: administrar rutas, definir nodos, elegir assets, editar indicaciones y publicar cambios.



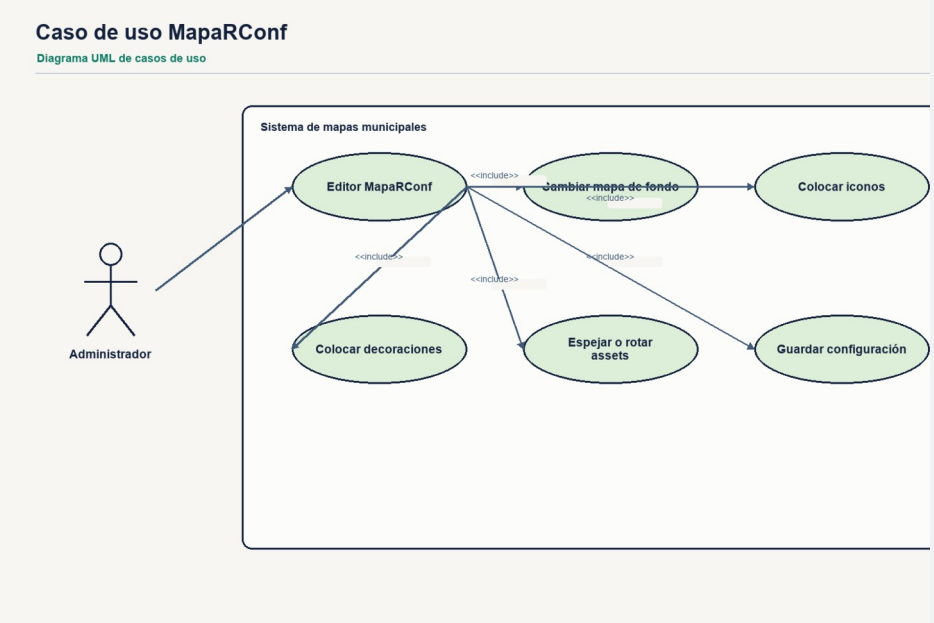

```
@startuml
title Caso de uso MapaConf
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

left to right direction
actor Administrador
rectangle "Editor MapaConf" {
  usecase "Crear, duplicar o
archivar ruta" as UC1
  usecase "Agregar, mover o
borrar nodos" as UC2
  usecase "Elegir asset
por tramo" as UC3
  usecase "Editar texto
y audio" as UC4
  usecase "Guardar
configuración" as UC5
}
Administrador --> UC1
Administrador --> UC2
Administrador --> UC4
UC1 .> UC3 : <<include>>
UC1 .> UC5 : <<include>>
@enduml
```

Figura 3. Caso de uso MapaConf

4. Caso de uso MapaRConf

Representa la administración del mapa urbano: rutas, fondo, iconos, decoraciones, transformaciones visuales y persistencia de cambios.




```
@startuml
title Caso de uso MapaRConf
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

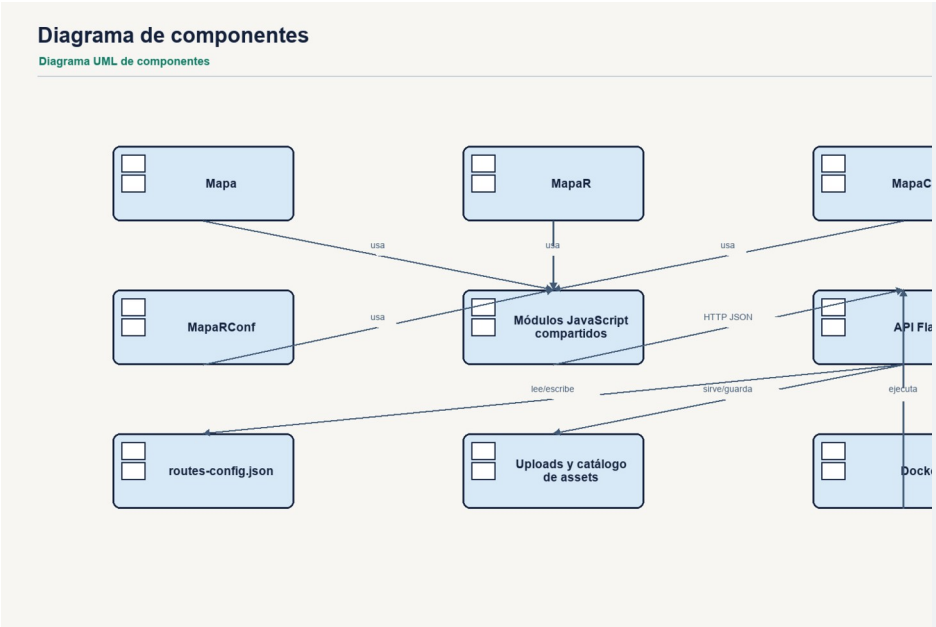
left to right direction
actor Administrador
rectangle "Editor MapaRConf" {
  usecase "Crear o editar
ruta urbana" as UC1
  usecase "Cambiar mapa
de fondo" as UC2
  usecase "Colocar iconos" as UC3
  usecase "Colocar
decoraciones" as UC4
  usecase "Rotar, espejar y
cambiar tamaño" as UC5
  usecase "Guardar
configuración" as UC6
}
Administrador --> UC1
Administrador --> UC2
Administrador --> UC3
Administrador --> UC4
UC3 .> UC5 : <<include>>
UC4 .> UC5 : <<include>>
UC1 .> UC6 : <<include>>
@enduml
```

Figura 4. Caso de uso MapaRConf

Componentes

5. Diagrama de componentes

Muestra la arquitectura: cuatro pantallas frontend consumen módulos compartidos y una API Flask, que persiste configuración y sirve los archivos subidos.




```
@startuml
title Diagrama de componentes
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

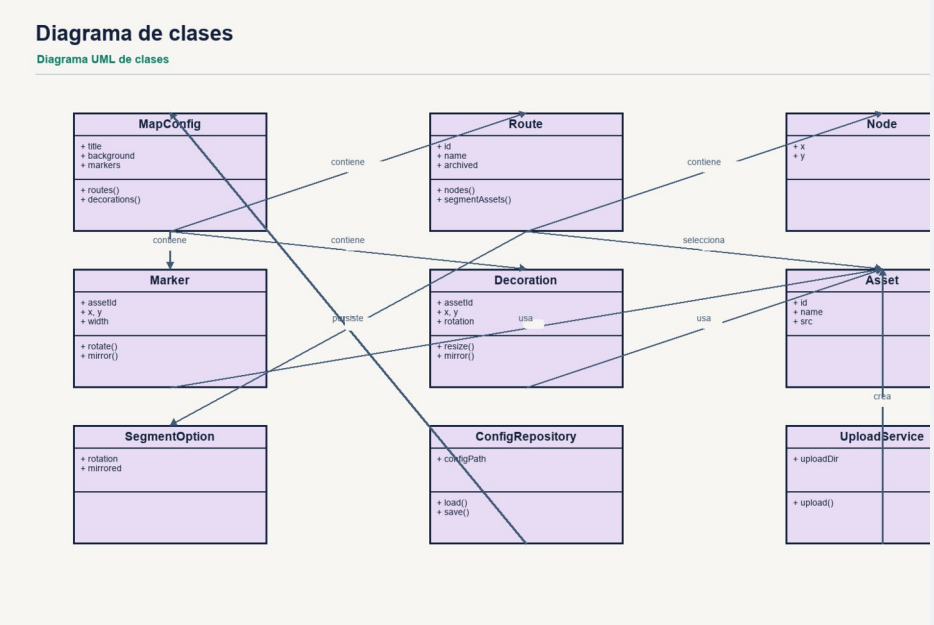
package "Frontend" {
  [Mapa] as mapa
  [MapaR] as mapar
  [MapaConf] as conf
  [MapaRConf] as maparconf
  [Módulos compartidos
config api map utils route viewer] as shared
}
component "API Flask" as api
database "routes config json" as config
folder "Uploads
mapas iconos audios
assets decoraciones" as uploads
node Docker as docker
mapa --> shared
mapar --> shared
conf --> shared
maparconf --> shared
shared --> api : HTTP JSON
api --> config : lee y guarda
api --> uploads : sirve y sube
docker --> api : ejecuta
@enduml
```

Figura 5. Diagrama de componentes

Clases

6. Diagrama de clases

Modela la estructura de datos compartida. Un mapa contiene rutas, marcadores y decoraciones; una ruta tiene nodos, assets por tramo y opciones de rotación o espejo.




```
@startuml
title Diagrama de clases
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

class MapConfig {
  +title: string
  +background: string
  +viewBox: string
}
class Route {
  +id: string
  +name: string
  +archived: boolean
  +instructions: string
  +audio: string
}
class Node { +x: number
+x: number }
class Marker { +assetId: string
+x: number
+y: number
+rotation: number
+mirrored: boolean }
class Decoration { +assetId: string
+x: number
+y: number
+width: number
+rotation: number }
class Asset { +id: string
+name: string
+src: string }
class SegmentOption { +rotation: number
+mirrored: boolean }
class ConfigRepository { +load()
+save() }
class UploadService { +upload(file, kind) }
MapConfig "1" *-- "0..*" Route
MapConfig "1" *-- "0..*" Marker
MapConfig "1" *-- "0..*" Decoration
Route "1" *-- "2..*" Node
Route "1" *-- "0..*" SegmentOption
Route ..> Asset : usa
Marker ..> Asset : usa
```

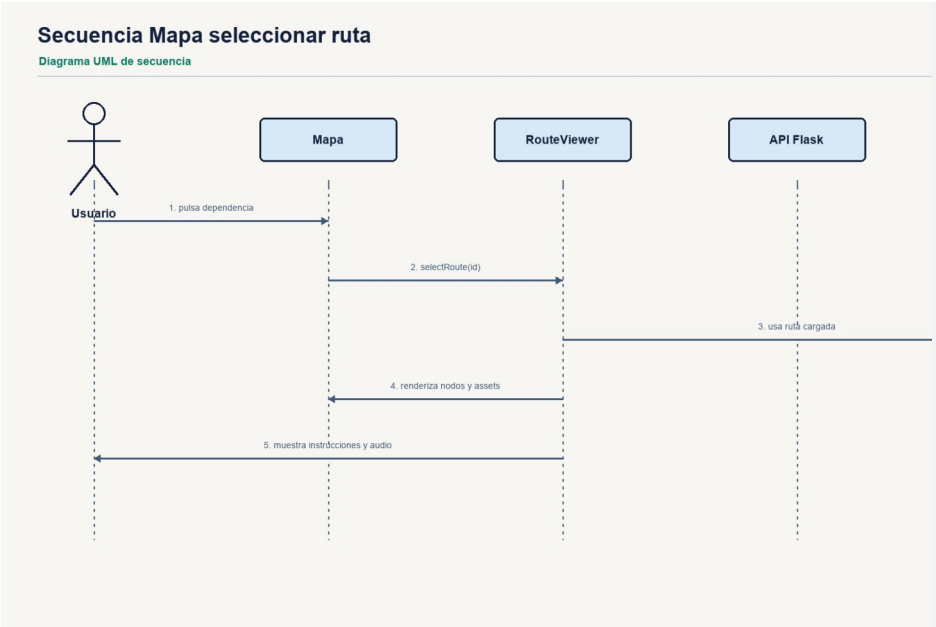
```
Decoration ..> Asset : usa
ConfigRepository --> MapConfig : persiste
UploadService --> Asset : crea
@enduml
```

Figura 6. Diagrama de clases

Secuencias

7. Secuencia Mapa seleccionar ruta

Explica el intercambio de mensajes cuando el visitante selecciona una dependencia del terminal y se dibuja el recorrido.



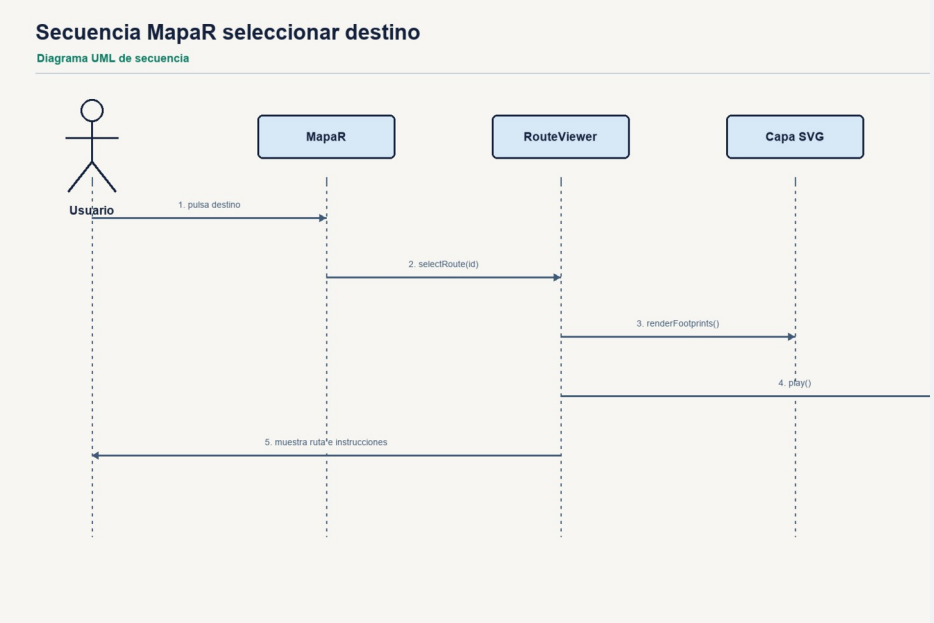

```
@startuml
title Secuencia Mapa seleccionar ruta
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

actor Usuario
participant Mapa
participant RouteViewer
database "Config JSON" as Config
participant Audio
Usuario -> Mapa : 1. pulsa dependencia
Mapa -> RouteViewer : 2. selectRoute(id)
RouteViewer -> Config : 3. obtiene ruta activa
Config --> RouteViewer : nodos assets instrucciones
RouteViewer -> Mapa : 4. renderiza recorrido SVG
RouteViewer -> Audio : 5. reproduce audio opcional
RouteViewer --> Usuario : 6. muestra pasos
@enduml
```

Figura 7. Secuencia Mapa seleccionar ruta

8. Secuencia MapaR seleccionar destino

Describe cómo MapaR procesa la selección de un destino, anima los assets y muestra indicaciones para el recorrido urbano.



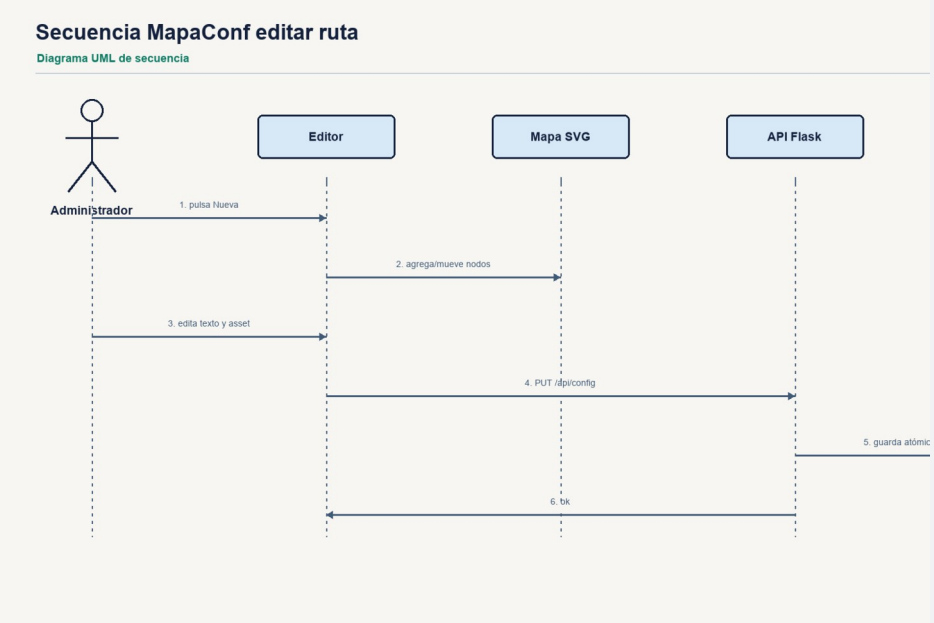

```
@startuml
title Secuencia MapaR seleccionar destino
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

actor Usuario
participant MapaR
participant RouteViewer
participant "Capa SVG" as SVG
participant Audio
Usuario -> MapaR : 1. pulsa destino
MapaR -> RouteViewer : 2. selectRoute(id)
RouteViewer -> SVG : 3. renderFootprints()
RouteViewer -> SVG : 4. renderMarkers()
RouteViewer -> Audio : 5. play()
RouteViewer --> Usuario : 6. muestra indicaciones
@enduml
```

Figura 8. Secuencia MapaR seleccionar destino

9. Secuencia MapaConf editar ruta

Muestra el ciclo de edición de una ruta del terminal desde la modificación visual hasta el guardado atómico de la configuración.



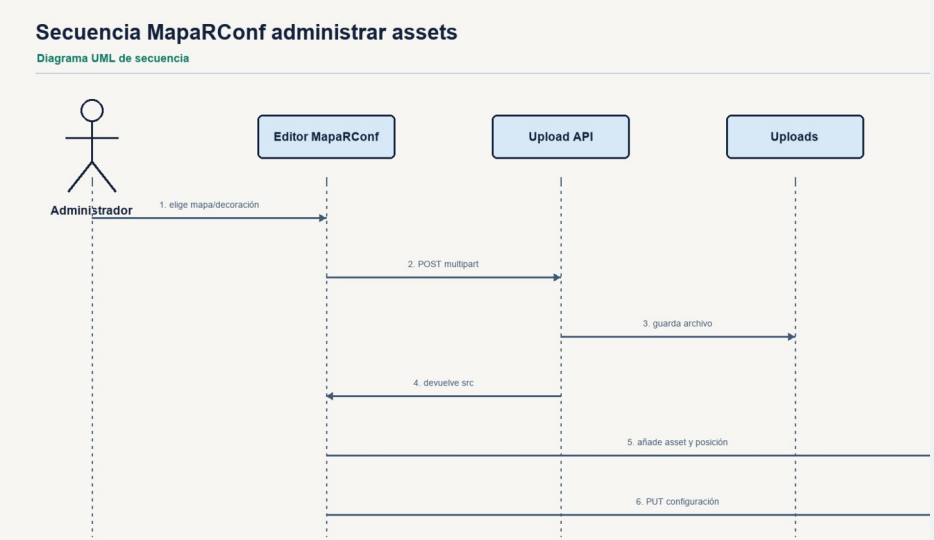

```
@startuml
title Secuencia MapaConf editar ruta
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

actor Administrador
participant RouteEditor
participant "Mapa SVG" as SVG
participant "API Flask" as API
database "routes config json" as Config
Administrador -> RouteEditor : 1. crea o selecciona ruta
Administrador -> RouteEditor : 2. modifica nodos assets texto
RouteEditor -> SVG : 3. actualiza vista previa
Administrador -> RouteEditor : 4. pulsa Guardar
RouteEditor -> API : 5. PUT api config
API -> Config : 6. guarda archivo temporal
API -> Config : 7. reemplaza config atómicamente
API --> RouteEditor : 8. confirmación
@enduml
```

Figura 9. Secuencia MapaConf editar ruta

10. Secuencia MapaRConf administrar assets

Explica la subida de un asset o mapa, su registro en el catálogo y la persistencia de su posición dentro del mapa urbano.




```
@startuml
title Secuencia MapaRConf administrar assets
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

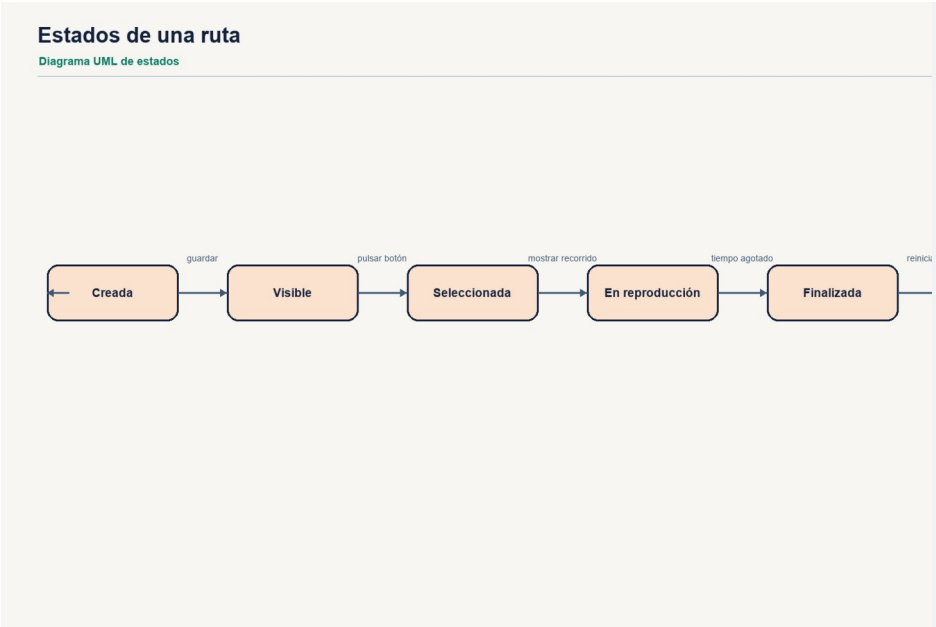
actor Administrador
participant RouteEditor
participant "Upload API" as API
folder Uploads
database "Config JSON" as Config
Administrador -> RouteEditor : 1. selecciona archivo
RouteEditor -> API : 2. POST api upload
API -> Uploads : 3. guarda archivo clasificado
API --> RouteEditor : 4. devuelve src y nombre
Administrador -> RouteEditor : 5. coloca y transforma asset
RouteEditor -> Config : 6. agrega asset y posición
RouteEditor -> API : 7. guarda configuración
@enduml
```

Figura 10. Secuencia MapaRConf administrar assets

Estados

11. Estados de una ruta

Explica el ciclo de vida de una ruta desde que se crea hasta que se archiva o vuelve a quedar visible.



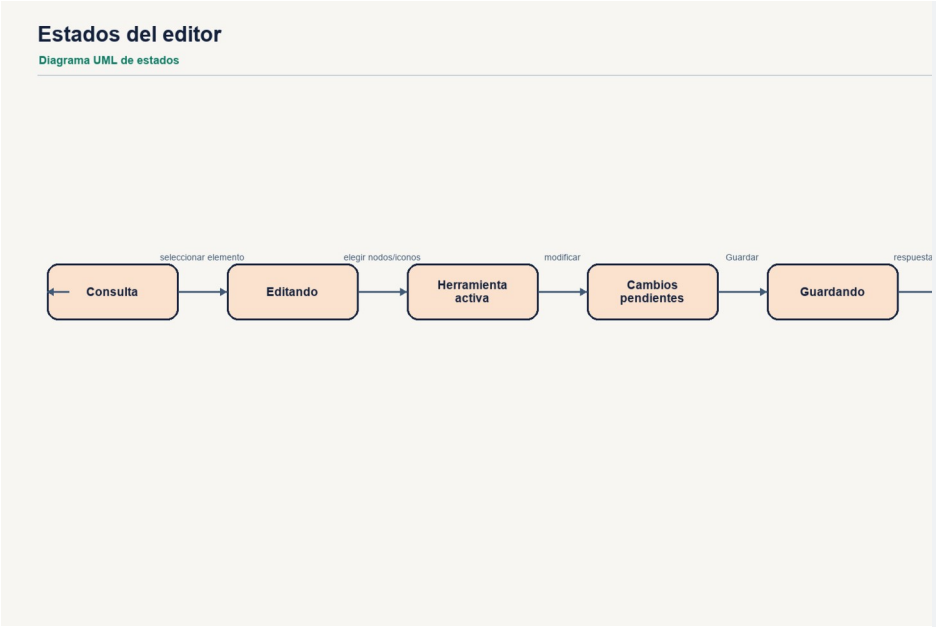

```
@startuml
title Estados de una ruta
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

[*] --> Creada
Creada --> Visible : guardar
Visible --> Seleccionada : usuario pulsa botón
Seleccionada --> MostrandoRecorrido : renderizar
MostrandoRecorrido --> Visible : temporizador termina
Visible --> Archivada : archivar
Archivada --> Visible : desarchivar
Visible --> [*] : eliminar
@enduml
```

Figura 11. Estados de una ruta

12. Estados del editor

Muestra los estados de trabajo del editor: selección, edición, cambios pendientes, guardado correcto o error de persistencia.



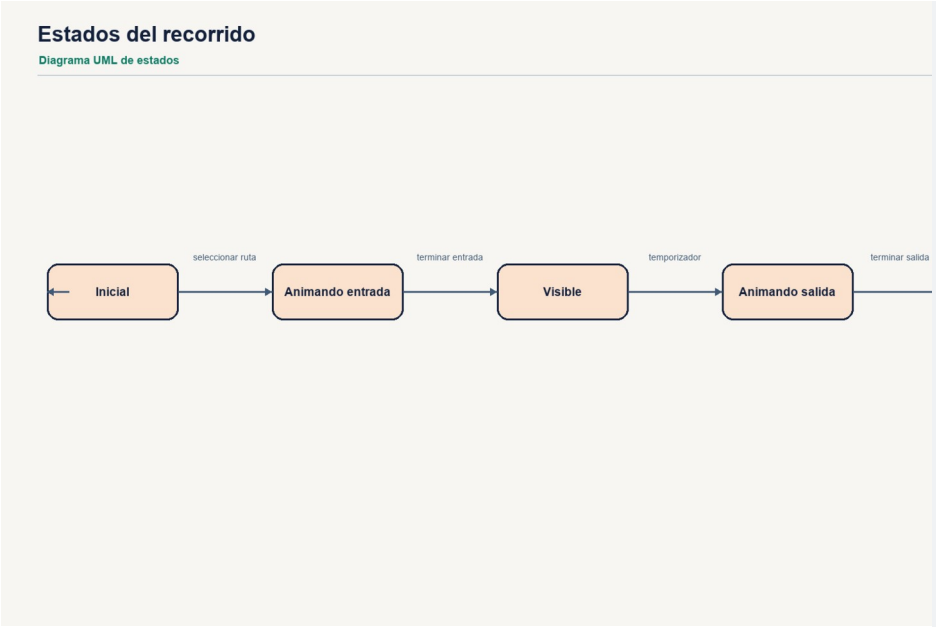

```
@startuml
title Estados del editor
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

[*] --> Consulta
Consulta --> Edicion : seleccionar ruta
Edicion --> HerramientaActiva : nodos iconos assets
HerramientaActiva --> CambiosPendientes : modificar
CambiosPendientes --> Guardando : Guardar
Guardando --> Guardado : respuesta 200
Guardando --> Error : error API
Error --> CambiosPendientes : corregir
Guardado --> Edicion : continuar
@enduml
```

Figura 12. Estados del editor

13. Estados del recorrido

Muestra el comportamiento visual del recorrido desde el panel inicial hasta la animación de salida y el reinicio.




```
@startuml
title Estados del recorrido
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

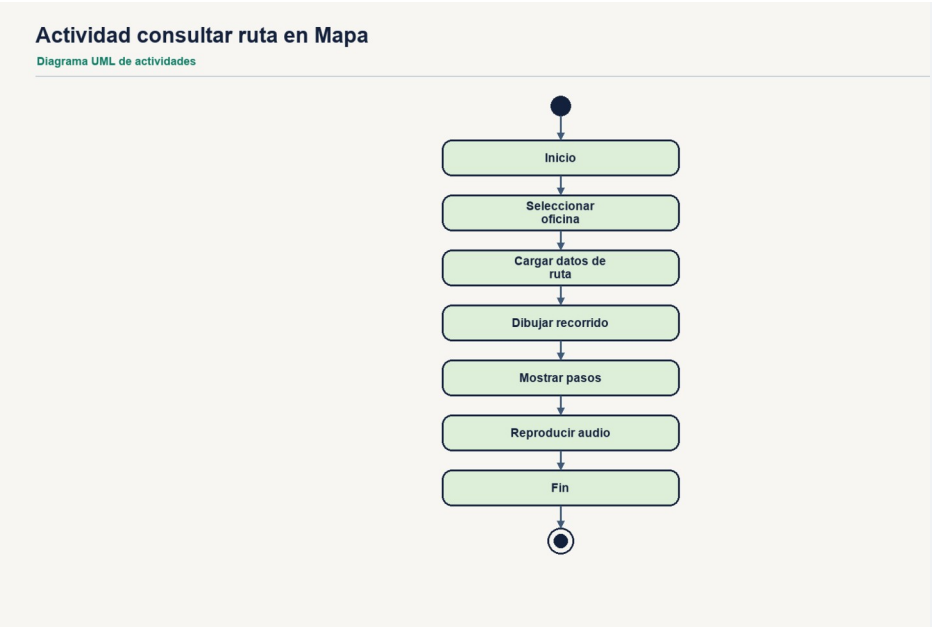
[*] --> PanelInicial
PanelInicial --> Entrada : seleccionar ruta
Entrada --> RutaVisible : animación completa
RutaVisible --> Salida : routeVisibleTime
Salida --> PanelInicial : animación completa
RutaVisible --> Entrada : seleccionar otra ruta
@enduml
```

Figura 13. Estados del recorrido

Actividades

14. Actividad consultar ruta en Mapa

Describe el flujo operativo del visor del terminal cuando un visitante consulta una oficina.



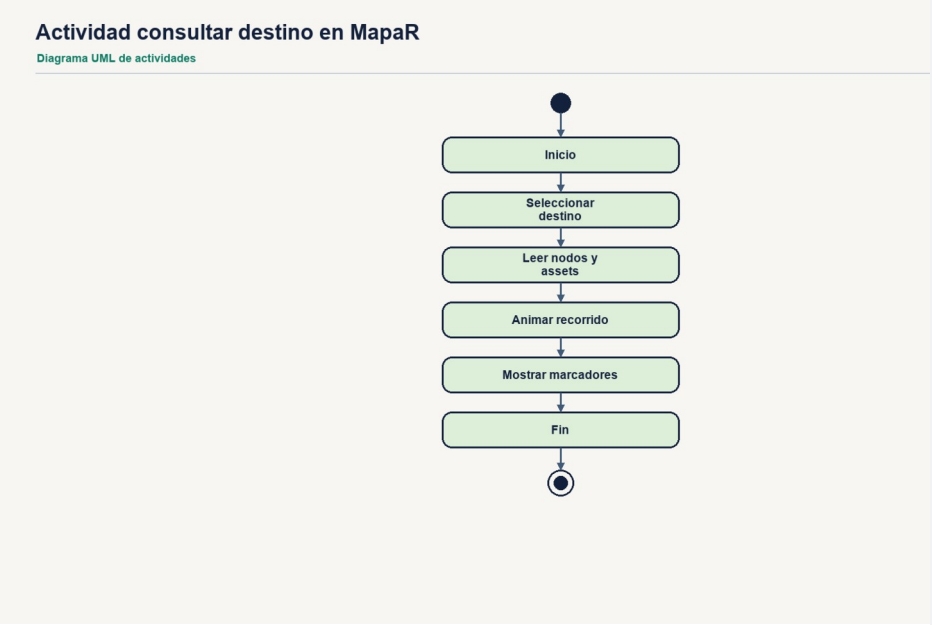

```
@startuml
title Actividad consultar ruta en Mapa
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

start
:Seleccionar oficina;
:Cargar nodos y assets;
:Dibujar recorrido;
:Mostrar pasos;
if (La ruta tiene audio) then (sí)
  :Reproducir audio;
endif
:Esperar temporizador;
:Ocultar recorrido;
stop
@enduml
```

Figura 14. Actividad consultar ruta en Mapa

15. Actividad consultar destino en MapaR

Describe la consulta de un destino urbano y la composición del recorrido con marcadores, assets e instrucciones.



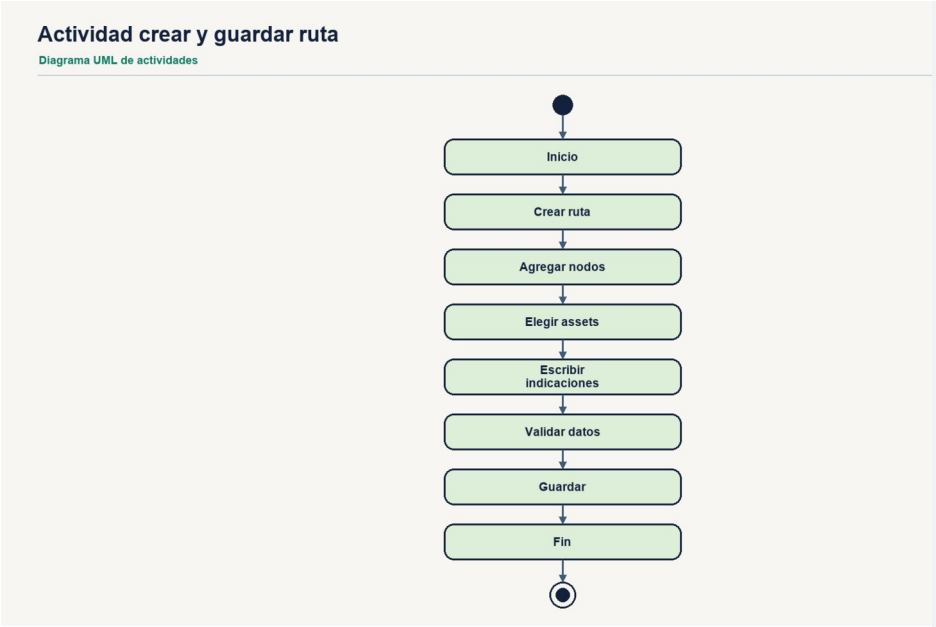

```
@startuml
title Actividad consultar destino en MapaR
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

start
:Seleccionar destino;
:Leer nodos y segment assets;
:Renderizar huellas o vehículo;
:Mostrar iconos y decoraciones;
:Mostrar indicaciones;
if (Hay audio) then (sí)
  :Reproducir audio;
endif
:Restablecer mapa;
stop
@enduml
```

Figura 15. Actividad consultar destino en MapaR

16. Actividad crear y guardar ruta

Muestra el proceso que sigue el administrador para publicar una ruta nueva desde cualquiera de los dos editores.



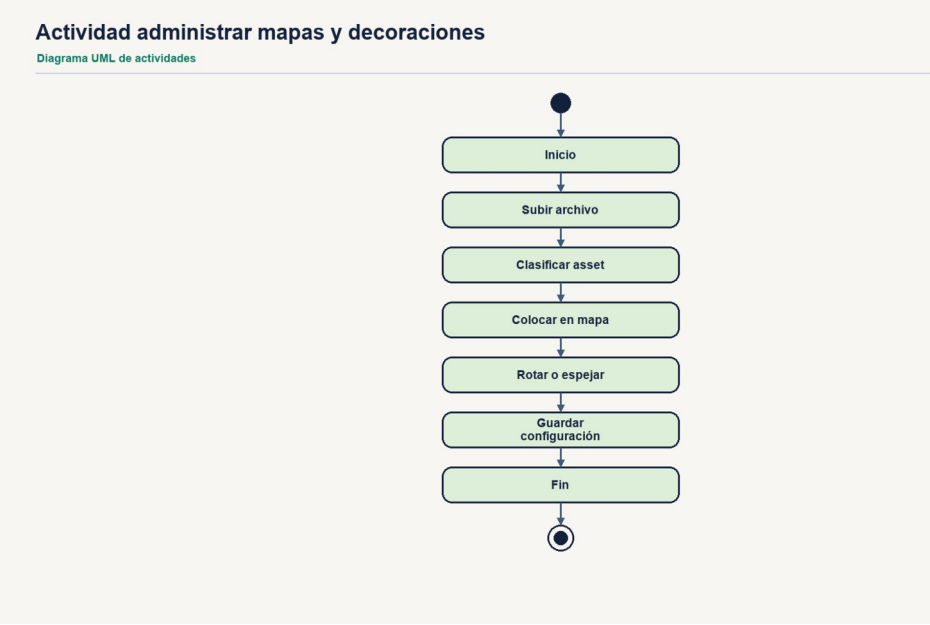

```
@startuml
title Actividad crear y guardar ruta
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

start
:Crear ruta;
:Definir nombre y color;
:Agregar nodos;
:Elegir assets por tramo;
:Escribir indicaciones;
if (Desea audio) then (sí)
  :Subir o seleccionar audio;
endif
:Guardar configuración;
:Ruta visible en el visor;
stop
@enduml
```

Figura 16. Actividad crear y guardar ruta

17. Actividad administrar mapas e iconos

Muestra cómo el administrador sube o selecciona un mapa, icono o decoración, aplica transformaciones y guarda el resultado.




```
@startuml
title Actividad administrar mapas e iconos
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

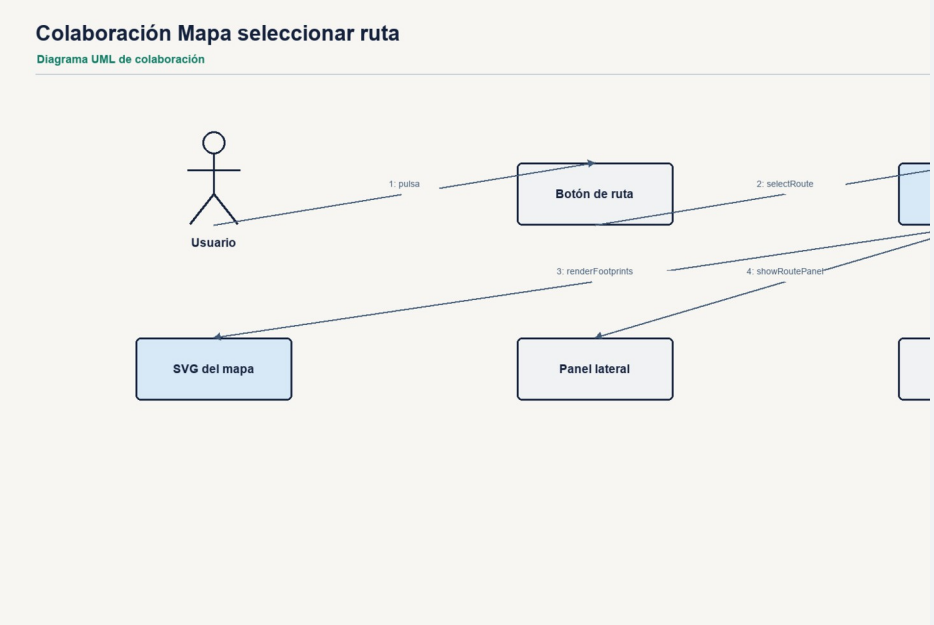
start
:Elegir modo mapa iconos o decoraciones;
:Seleccionar o subir archivo;
:Clasificar asset;
:Colocar sobre el mapa;
:Mover y cambiar tamaño;
:Rotar o espejar;
:Guardar configuración;
stop
@enduml
```

Figura 17. Actividad administrar mapas e iconos

Colaboración

18. Colaboración Mapa seleccionar ruta

Muestra los objetos que cooperan durante la selección de una ruta normal, incluyendo interfaz, visor, SVG, panel y audio.



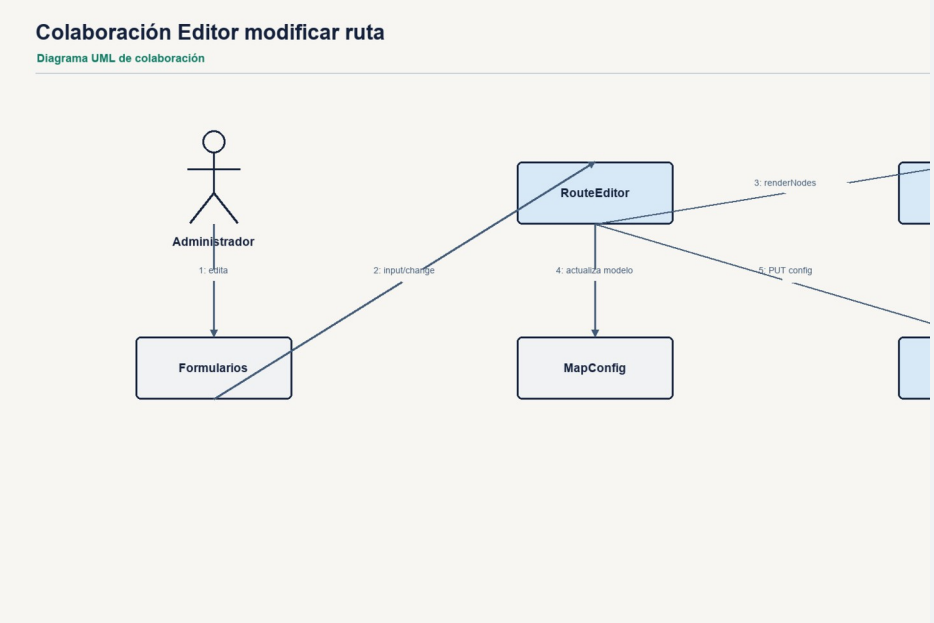

```
@startuml
title Colaboración Mapa seleccionar ruta
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

object Usuario
object "Botón de ruta" as Boton
object RouteViewer
object "Capa SVG" as SVG
object "Panel lateral" as Panel
object Audio
Usuario --> Boton : 1 pulsa
Boton --> RouteViewer : 2 selectRoute
RouteViewer --> SVG : 3 renderFootprints
RouteViewer --> Panel : 4 showRoutePanel
RouteViewer --> Audio : 5 playRouteAudio
@enduml
```

Figura 18. Colaboración Mapa seleccionar ruta

19. Colaboración Editor modificar ruta

Muestra la cooperación entre administrador, formularios, editor, modelo de configuración, mapa SVG y la API de guardado.



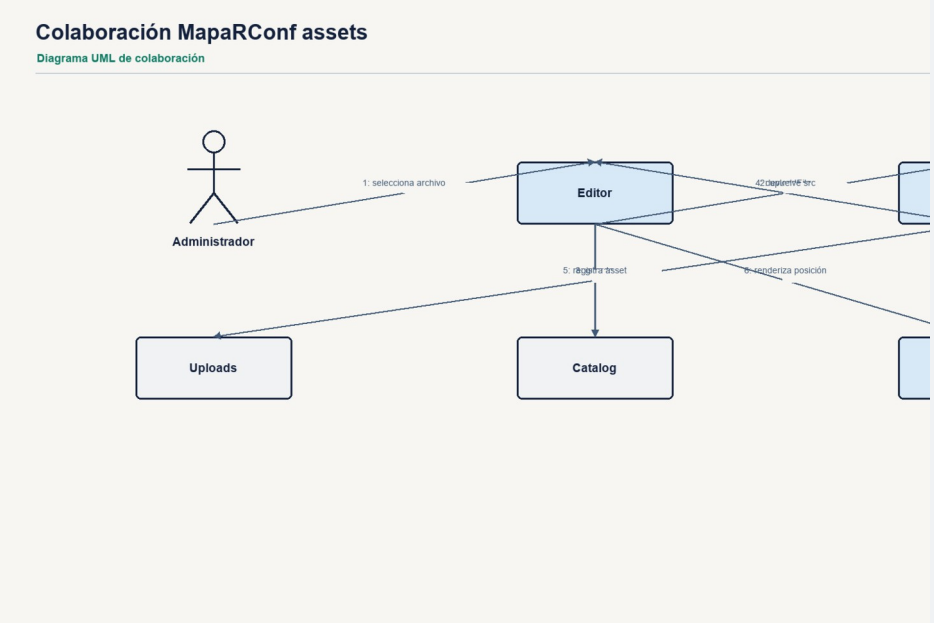

```
@startuml
title Colaboración Editor modificar ruta
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

object Administrador
object Formularios
object RouteEditor
object "Mapa SVG" as SVG
object MapConfig
object "Save API" as API
Administrador --> Formularios : 1 edita
Formularios --> RouteEditor : 2 input change
RouteEditor --> SVG : 3 renderNodes
RouteEditor --> MapConfig : 4 actualiza modelo
RouteEditor --> API : 5 PUT config
@enduml
```

Figura 19. Colaboración Editor modificar ruta

20. Colaboración MapaRConf assets

Muestra el intercambio entre editor, API de uploads, catálogo y capa del mapa cuando se sube o coloca un asset.




```
@startuml
title Colaboración MapaRConf assets
!theme plain
skinparam backgroundColor #F8F7F3
skinparam shadowing false
skinparam defaultFontName Arial
skinparam defaultFontColor #14213D
skinparam ArrowColor #415A77
skinparam BorderColor #14213D
skinparam packageBorderColor #14213D
skinparam packageBackgroundColor #FFFFFF
skinparam usecaseBackgroundColor #DDEFD9
skinparam usecaseBorderColor #14213D
skinparam componentBackgroundColor #D7E8F7
skinparam classBackgroundColor #E7DCF3
skinparam stateBackgroundColor #FBE2CF
skinparam activityBackgroundColor #DDEFD9
skinparam objectBackgroundColor #F0F2F4
skinparam sequenceParticipantBackgroundColor #D7E8F7
skinparam sequenceLifeLineBorderColor #415A77
skinparam sequenceArrowColor #415A77
skinparam roundcorner 12

object Administrador
object Editor
object "Upload API" as API
object Uploads
object Catalogo
object "Map Layer" as MapLayer
Administrador --> Editor : 1 selecciona archivo
Editor --> API : 2 uploadFile
API --> Uploads : 3 guarda archivo
API --> Editor : 4 devuelve src
Editor --> Catalogo : 5 registra asset
Editor --> MapLayer : 6 renderiza posición
@enduml
```

Figura 20. Colaboración MapaRConf assets